

Today's Content :-

Welcome everyone 😊

- a) Class & Object Concept
- b) Object & Object Reference
- c) Object Reference as Member

24th Dec - 3rd Jan.

Class :- It is a blueprint  blueprint

Object :- Real Instance of blueprint 
creation

Class :-
 → Attributes : information | Properties
 → functionaliti : action

class car {

colour
 seater
 brand
 break()
 airbag()
 automanual()
 speed()
 }

[examples of objects]

Shrutis car

colour: black
 seater: 4
 brand: 
 break()
 airbag()
 automanual()
 speed()

Deepthi's car

colour: Red
 seater: 7
 brand: 
 break()
 airbag()
 automanual()
 speed()

Student Class:-

class Student {

Attributes {
 int rollno;
 String name;
 int m1, m2, m3;

function {
 int totalmarks() {
 return m1+m2+m3;
 }
 int max-marks() {
 return Math.max(m1, Math.max(m2,m3));
 }
 void printname() {
 Print(name)
 }
 }

classname st1 = new classname();

{ Student st1 = new Student();

st1 →
 rollno = 65
 name = Deepika
 m1 = 65
 m2 = 99
 m3 = 98

We need . operator st1.m3=98

st1.rollno = 65 st1.m1 = 77
 st1.name = "Deepika" st1.m2 = 99

Object & Object Reference Note:- Object Reference of class can hold

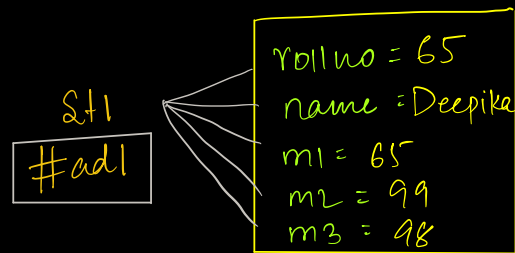
class Student {

```
int rollno;  
String name;  
int m1, m2, m3;  
int totalmarks() {  
}  
int max-marks() {  
}  
void printname() {  
}  
}
```

Student st1 = new Student();

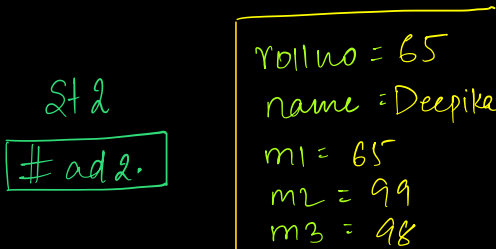
Object Reference
of class Student

this line will
create object



#ad1
↳ address of above
object

Student st2 = new Student();



#ad2
address of above
object

Student st3 = null;

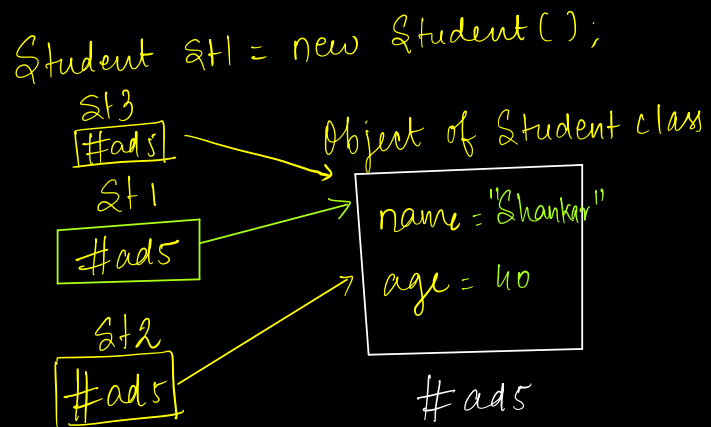
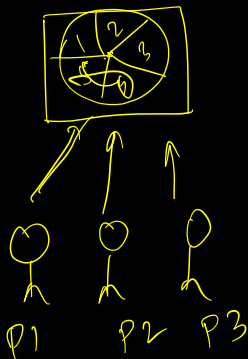
#null

st3.m1 = 50 } null pointer exception.

(Multiple Object References)

```
class Student {  
    String name;  
    int age;  
    Study()  
    Drink()  
    Falllove()  
}
```

```
{  
    int x = 5;  
    int y = x;  
}
```



Student st2 = st1;



Student st3 = st2;



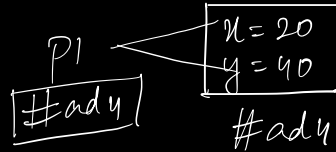
```

class Pair {
    int x, y;
}

```

Pair p1 = new Pair() // Create object

p1.x = 20
p2.y = 40 } to initialize attributes



Constructor :-

↳ It is used to initialize attributes of class, at the time of object creation itself.

```

class Pair {

```

a) Constructor name should be same as class name

```

    int x, y;

```

b) It is similar to function but no return type

```

    Pair(int a, int b) {

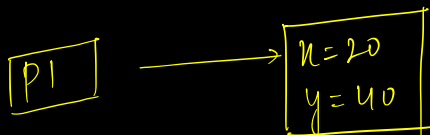
```

```

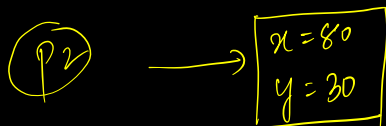
        x = a;
        y = b;
    }
}

```

Pair p1 = new Pair(20, 40);



Pair p2 = new Pair(80, 30);



// Basics of Objects & Classes

- Tree Basics
- Linked list Basics

Use this

[illegible]

Sat → PS session - 9pm }